# Document Control

**Title**

**Patients receiving dabigatran requiring emergency reversal for surgery or treatment of haemorrhage Guidelines**

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|---|---|
| **Author** | **Author's job title**<br>Pharmacist |
| **Directorate**<br>PCS - Pharmacy | **Department**<br>Pharmacy |

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|---|---|
| **Main Contact**<br>Anticoagulation Pharmacy Lead<br>North Devon District Hospital<br>Raleigh Park<br>Barnstaple, EX31 4JB | **Tel: Direct Dial** – 01271 322395 |

**Lead Director**
Director of Pharmacy

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**Consulted with the following stakeholders: (list all)**

- Emergency Medicine
- Anaesthetics
- Haematology

**Approval and Review Process**

- Drug and Therapeutics
- Anticoagulation Working Group

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## 1. Purpose

1.1. The purpose of this document is to detail the process for using idarucizumab (Praxbind) for emergency reversal of dabigatran.

1.2. The policy applies to all appropriate Trust staff.

1.3. Implementation of this policy will ensure that:

- Idarucizumab is used safely and appropriately

## 2. Definitions

2.1. NOAC – novel oral anticoagulants

2.2. Idarucizumab (Praxbind®) – reversal agent for dabigatran

## 3. Responsibilities

3.1 Is it the responsibility of all parties involved to ensure idarucizumab is used safely and appropriately.

The decision to prescribe idarucizumab should be made by a consultant or middle grade where a consultant is not available.

## 4. Background

Apixaban, dabigatran, rivaroxaban and edoxaban are direct oral anticoagulants that are alternatives to coumarins (e.g. warfarin) in selected groups of patients for certain indications. This guideline is for patients receiving dabigatran (Pradaxa®) requiring rapid reversal for an invasive procedure, emergency surgery or treatment of uncontrollable haemorrhage.

Dabigatran is a direct factor thrombin inhibitor licensed to prevent stroke and systemic embolism in adult patients with non-valvular atrial fibrillation plus the treatment of deep vein thrombosis (DVT) and pulmonary embolism (PE), and prevention of recurrent DVT and PE in adults.

Dabigatran does not routinely require monitoring of therapeutic response (unlike warfarin).

However, if a patient has an episode of bleeding or requires an invasive procedure, measurement of an anticoagulant effect may be advantageous.

The effect of dabigatran on routine coagulation screening tests is dependent on factors specific to the laboratory performing the test. The aPTT is the most useful test for assessing the anticoagulant effect of dabigatran. A normal aPTT is likely to exclude a therapeutic level of anticoagulation with dabigatran. A standard thrombin time is likely to be very sensitive to dabigatran, but is not recommended for assessment of level of anticoagulation.

**Appendix 3** shows the effect of all the direct oral anticoagulants on clotting screens

## Emergency Surgery

Dabigatran is usually stopped in advance of surgery and delaying surgery until no longer anticoagulated is preferential – see trust guideline on anticoagulants and surgery.

If the procedure cannot be delayed for the time required, the increased risk of bleeding should be assessed against the urgency of the intervention. Dabigatran should be re-started post procedure when risk of bleeding is judged to be low. The timing of reintroduction will depend on the clinical situation and it may be preferable to use standard anticoagulants such as enoxaparin

**Emergency surgery –** see **Appendix 1** flowchart below

## Reversal or overdose

**Praxbind (idarucizumab) is a specific reversal agent for dabigatran in adults when rapid reversal is required**. The dose is 5g (2x2.5g vials) given as two consecutive infusions over 5 to 10 minutes each or as a bolus injection. It is kept in the emergency department and should be used only where absolutely necessary. The vials should be kept in the fridge protected from light and used within an hour of opening. If the patient has bleeding complications (related to overdose or otherwise) please see **Appendix 2** flowchart.

# 5. Monitoring Compliance with and the Effectiveness of the Guideline

## Standards/ Key Performance Indicators

5.1. Annual audit of idarucizumab usage

## Process for Implementation and Monitoring Compliance and Effectiveness

5.2. Idarucizumab is already available in ED.

5.3. Training to be done through ED training days.

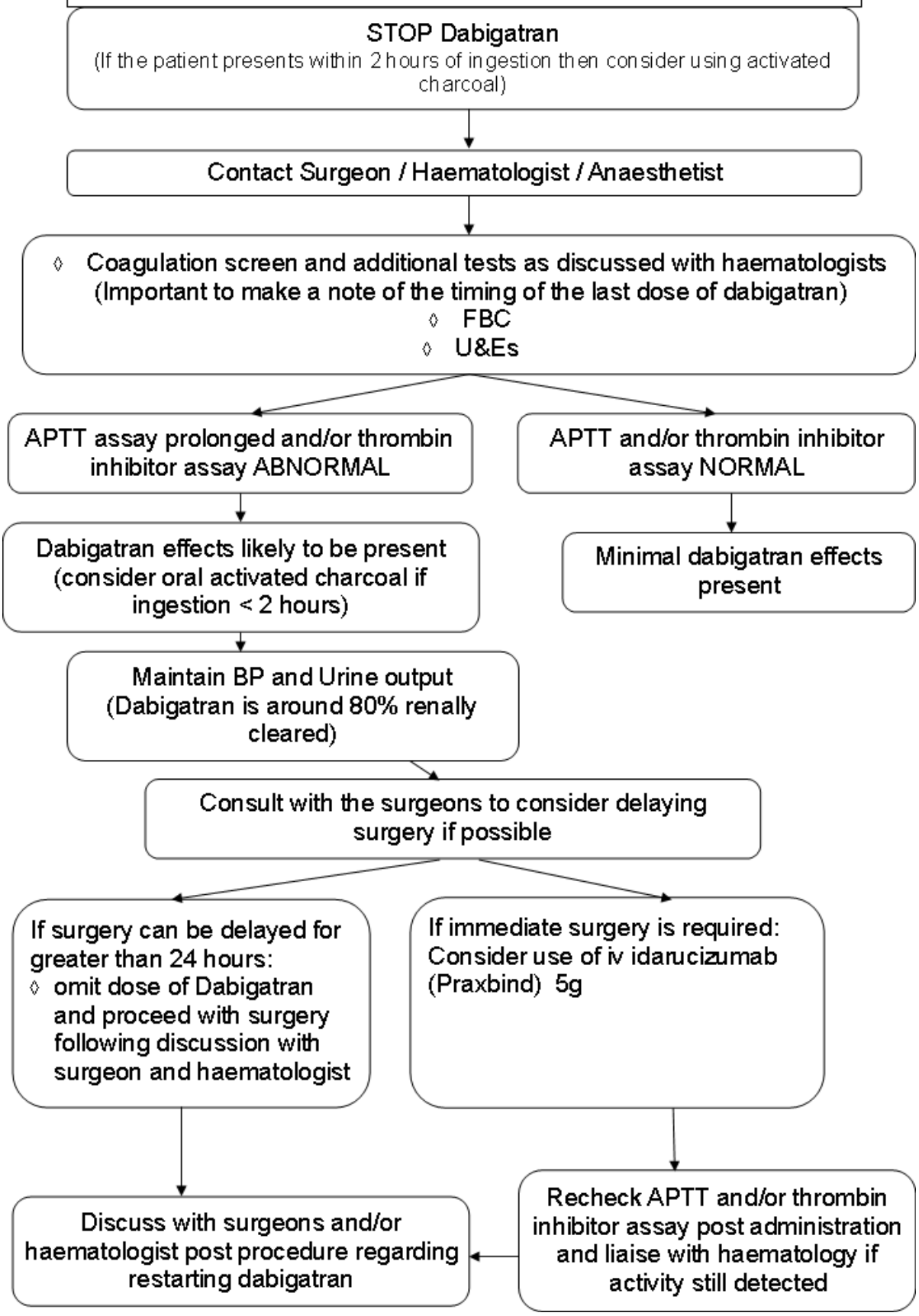
# 6. References


- Dabigatran (Pradaxa) Summary of Product Characteristics. Last Accessed 7/7/16 [https://www.medicines.org.uk/emc/medicine/20760]
- Idarucizumab (Praxbind) Summary of Product Characteristics Last Accessed 7/7/16 [https://www.medicines.org.uk/emc/medicine/31243]
- Based on guideline kindly provided by Julian Holmes at NUH NHS Trust.

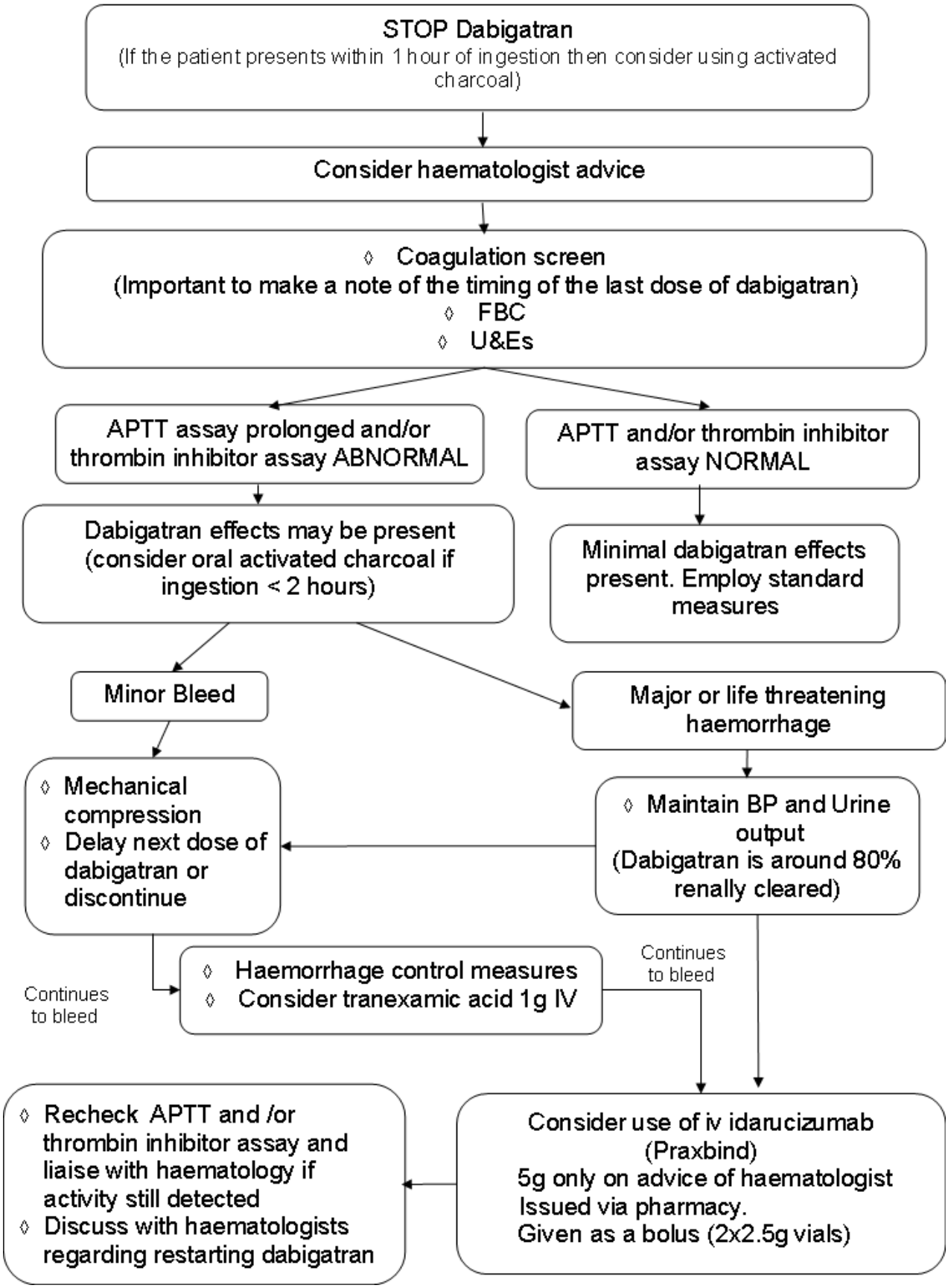

# 7. Associated Documentation

- Anticoagulation Policy
- Hatton-Wyatt, E. Pruchniewicz, J. (2016) Peri-operative management of patients on warfarin and the new oral anticoagulants. *Update in Anaesthesia. 31, 14-23.*

# **Appendix 1** Patient Receiving Dabigatran (Pradaxa®) Therapy – Emergency Surgery Protocol



# **Appendix 2** Patient Receiving Dabigatran (Pradaxa®) Therapy – Haemorrhage Protocol



# Appendix 3

Apixaban, dabigatran, rivaroxaban and edoxaban are direct oral anticoagulants that are alternatives to coumarins (e.g. warfarin) in selected groups of patients for certain indications. All these drugs accumulate in renal impairment. A standard clotting screen has not been validated for assessing the degree of anticoagulation in a patient taking these agents and should not be used for this purpose. Consult haematology for advice.

The table below gives information on the drugs' effects on coagulation screens:

| Parameter | Apixaban (Eliquis®) | Dabigatran (Pradaxa®) | Rivaroxaban (Xarelto®) | Edoxaban (Lixiana®) |
|---|---|---|---|---|
| **PT** | Prolonged | No effect | Prolonged (in linear fashion if neoplastin used as reagent) | Prolonged |
| **APTT** | Prolonged | Prolonged (1.4-1.8 times control) greatly prolonged if supratherapeutic levels | Prolonged (1.5-1.8 times control) | Prolonged |
| **TT** | No effect | Prolonged | No effect | No effect |
| **Drug Activity** | Use anti Xa assay | Use Haemoclot thrombin inhibitor assay or ECT | Use anti Xa assay | Use anti Xa assay |
| **Platelet count** | No effect | No effect | No effect | No effect |
| **D-dimer** | Suppressed levels | Suppressed levels | Suppressed levels | Suppressed levels |
| **Fibrinogen** | No effect | Can give falsely low results | No effect | No effect |